Assignment 1B

Cloud Computing Architecture
COS20019

Name: Pattarapol Tantechasa
Student ID: 103883220
Tutorial Session: Tuesday 10:30AM – 12:30AM
Submission Date: 14th April 2024

©2024 Pattarapol Tantechasa, Student ID: 103883220, COS20019

TABLE OF CONTENTS

I. Introduction …….. 2
II. Infrastructure Implementation ……. 3
A. Virtual Private Cloud ……. 3
B. Routing Tables …….. 3
C. Security Groups …… 4
a. TestInstanceSG …… 4
b. WebServerSG …….. 5
c. DBServerSG …… 5
D. Network ACLs ……. 6
E. EC2 Instances …….. 7
a. Bastion/Web Server (Public subnet 2) …… 7
b. Test Instance (Private subnet 2) …….. 7
c. Test ICMP Communication …… 8
d. Test SSH Communication …….. 8
F. RDS …….. 9
a. Database (Private subnet 1) …… 9
b. PhpMyAdmin ……... 10
G. S3 Bucket …… 11
III. Functionality …… 12
A. URL …… 12
B. Webpage (album.php) …… 12

I. INTRODUCTION

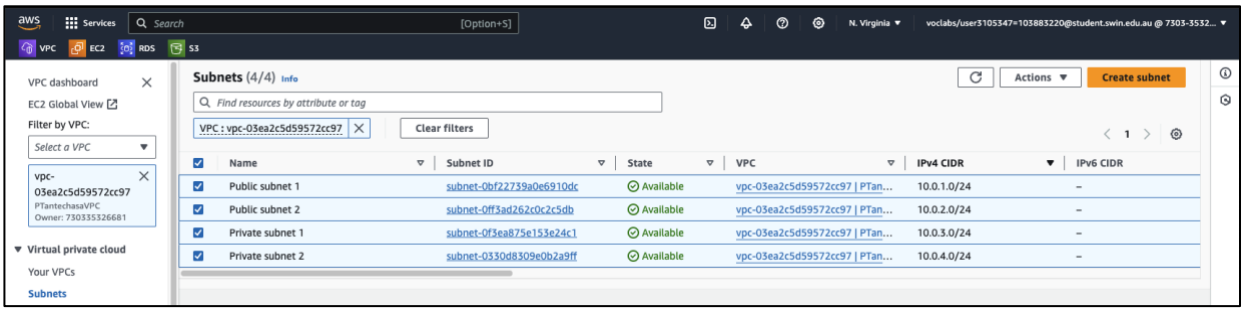
This report outlines the creation of secure Virtual Private Cloud (VPC) within the Amazon Web Services (AWS) ecosystem. The project covers the implementation and deployment of subnets, routing tables, security groups and setting up controlled entry through an Internet Gateway (IGW). The main objective is to guarantee maximum security and segregation of resources within the VPC.

Additionally, this project requires adjustment of a provided PHP code to build a dynamic website capable of storing metadata information from photos uploaded via Amazon Simple Storage Service (S3) into a MySQL database. The website will empower users to search for and showcase photos based on their associated metadata. To verify the infrastructure and functionality, the PHP website is deployed and thoroughly tested on an Apache web server operating on an Amazon Elastic Compute Cloud (EC2) virtual machine instance.

In conclusion, an extra layer of security is introduced by implementing Network Access Control Lists (NACLs) to fortify the protection of the public subnet housing the web server. This report details the entire process, starting from the initial VPC design to the successful deployment of a secure and fully functional web application.

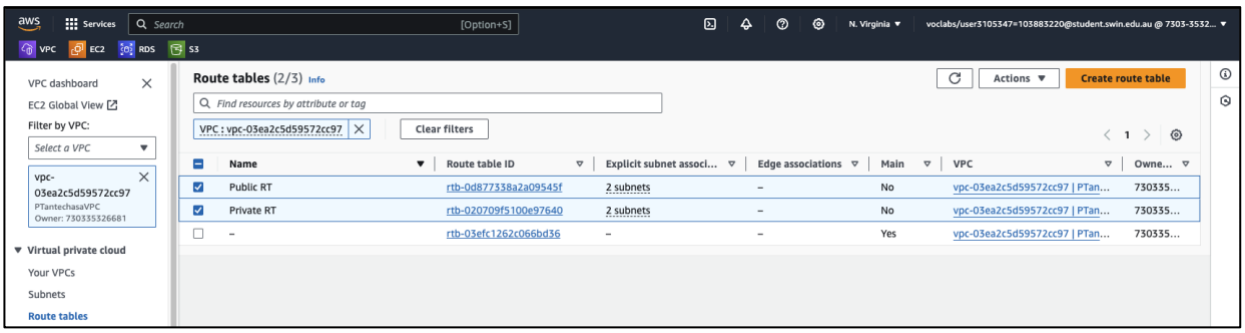
II. Infrastructure Implementation

A. *Virtual private cloud (VPC)*



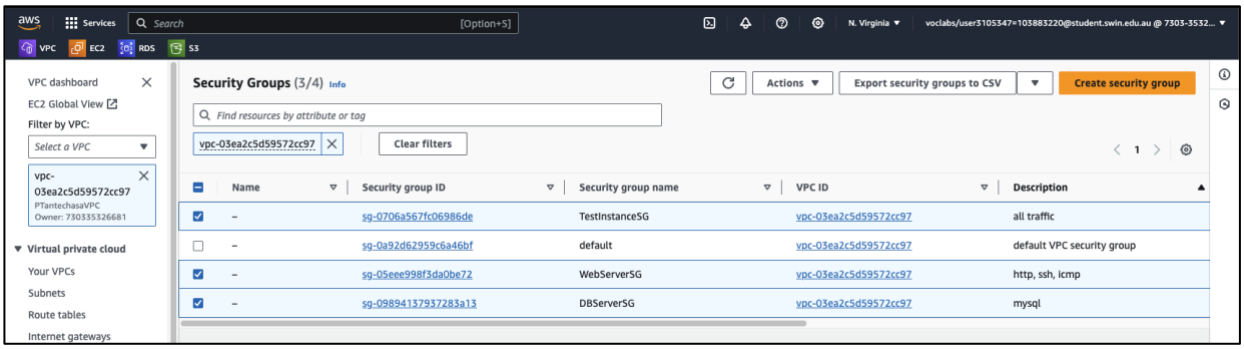
Name	*IPv4 CIDR*	*Availability Zone*
Public Subnet 1	10.0.1.0/24	US-EAST-1A
Private Subnet 1	10.0.3.0/24	US-EAST-1A
Public Subnet 2	10.0.2.0/24	US-EAST-1B
Private Subnet 2	10.0.4.0/24	US-EAST-1B

B. *Routing Tables*



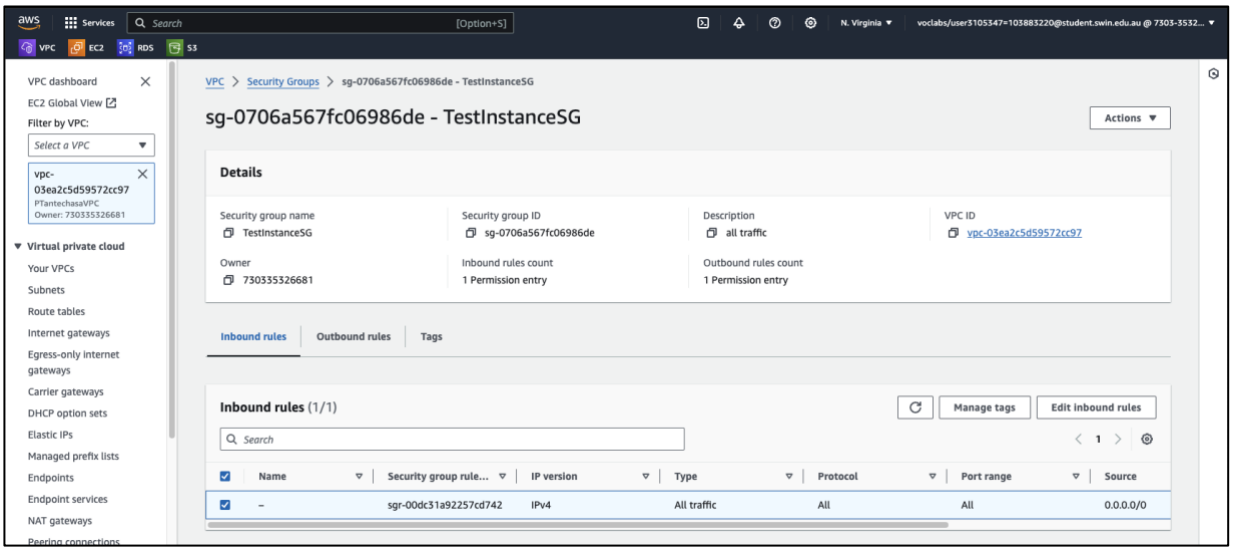
Name	*Explicit Subnet Associations*
Public RT	- Public Subnet 1 - Public Subnet 2
Private RT	- Private Subnet 1 - Private Subnet 2

C. Security Groups



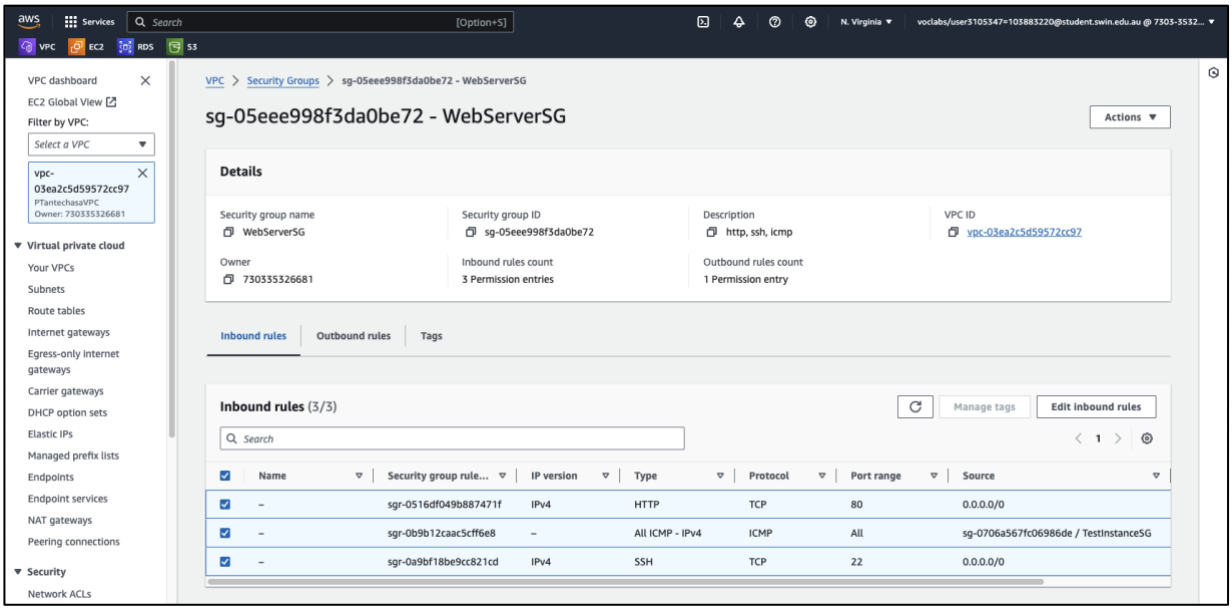
Here is an overview of all security groups within the VPC, with further details provided for each security group below.

a. TestInstanceSG



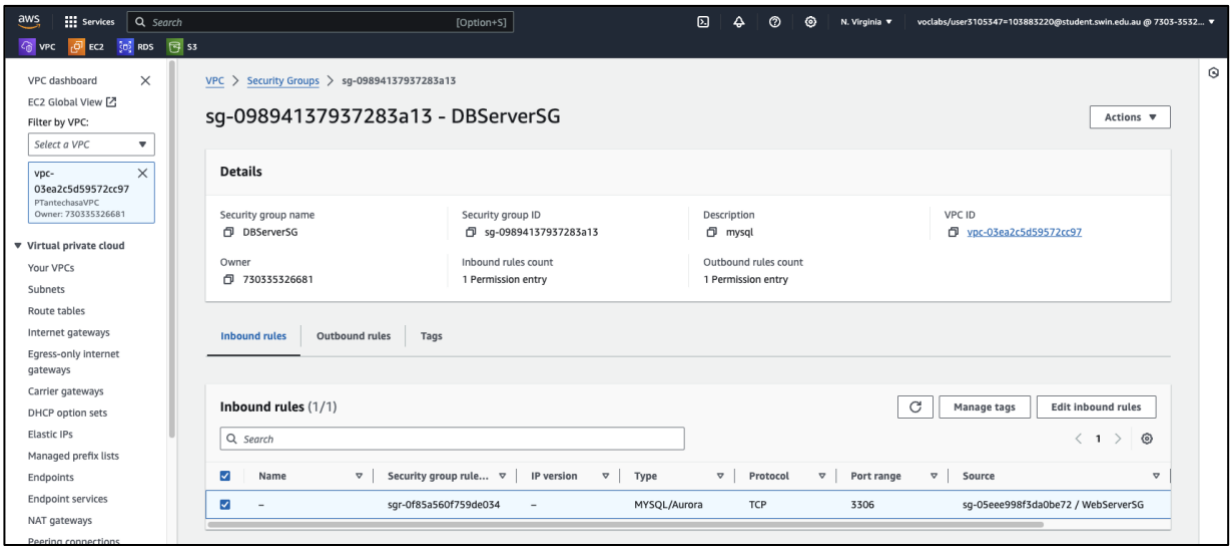
This security group is created for a "Test Instance" EC2 instance, its only allow all traffic from anywhere via IPv4.

b. WebServerSG



This security group is created for a "Web Server" EC2 instance, it allows HTTP and SSH communication from anywhere via IPv4. It also allows all ICMP communication via IPv4 only from "TestInstanceSG" security group.

c. DBServerSG

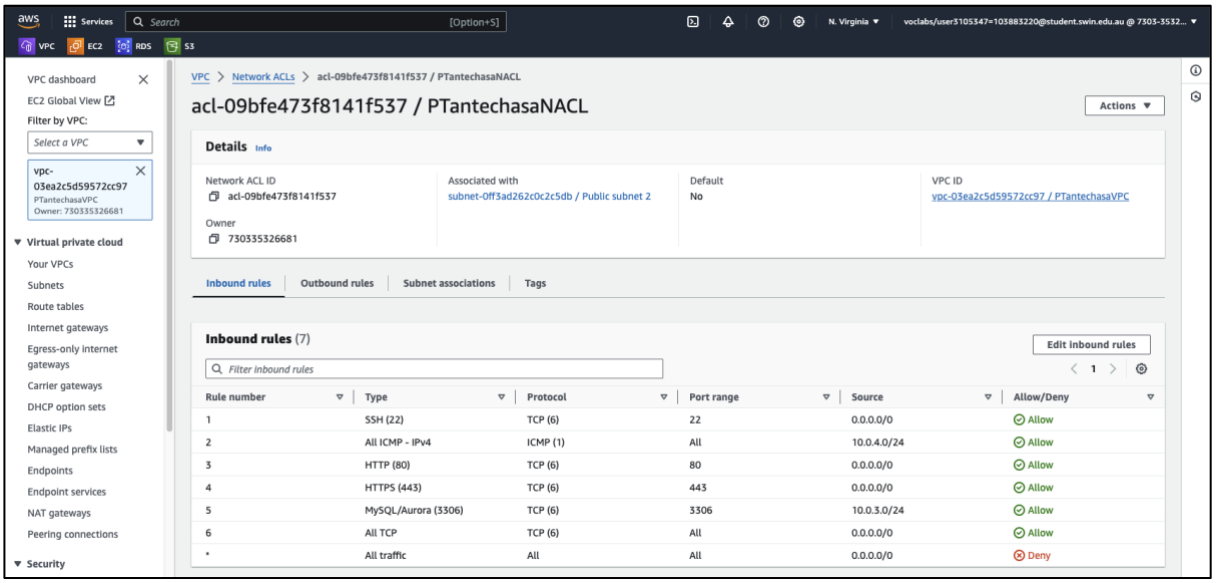


This security group is created for a database instance, it allows MySQL communication only from "WebServerSG" security group.

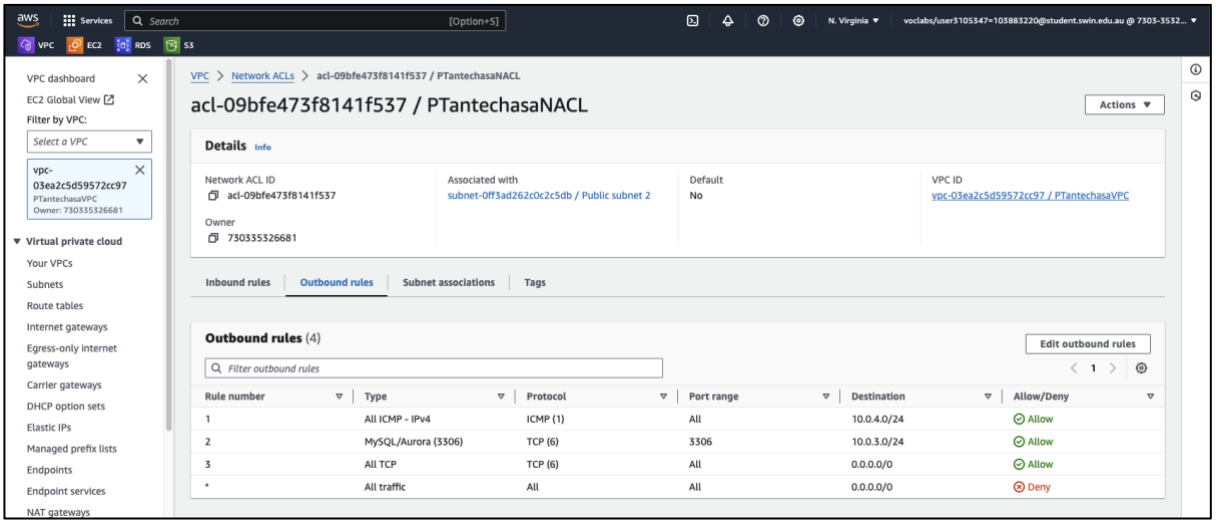
D. Network ACL

The Network Access Control List (NACL) was implemented to regulate both inbound and outbound communication for Public Subnet 2. Additional information regarding the specific inbound and outbound rules can be found below.

Inbound Rules

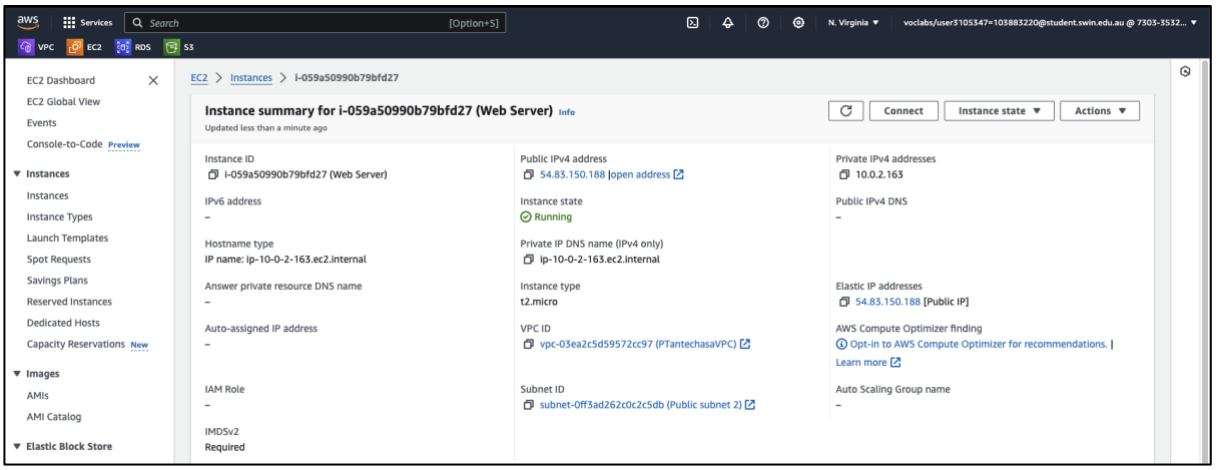


Outbound Rules



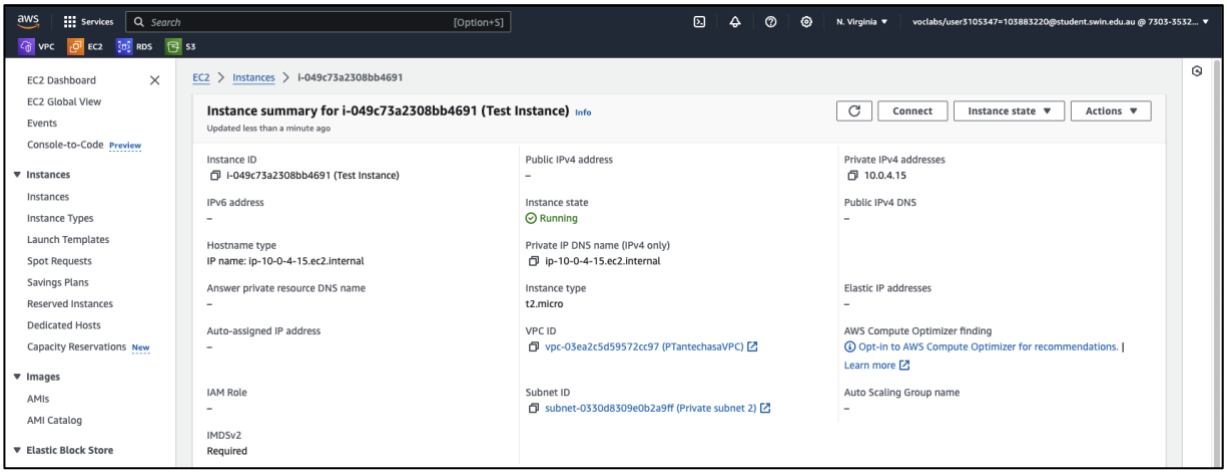
E. EC2 Instance

a. Web Server (Public Subnet 2)



This instance serves as a bastion host, providing users with SSH access to the Test Instance situated within the private subnet. An Elastic IP Address has been generated and assigned to this instance.

b. Test Instance (Private Subnet 2)



This instance is used for user to ping the web server using the "ping" command in Linux.

c. Test ICMP Communication

```
ec2-user@ip-10-0-2-163:~
[ec2-user@ip-10-0-2-163 ~]$ ping 10.0.4.15
PING 10.0.4.15 (10.0.4.15) 56(84) bytes of data.
64 bytes from 10.0.4.15: icmp_seq=1 ttl=127 time=0.460 ms
64 bytes from 10.0.4.15: icmp_seq=2 ttl=127 time=0.473 ms
64 bytes from 10.0.4.15: icmp_seq=3 ttl=127 time=0.485 ms
64 bytes from 10.0.4.15: icmp_seq=4 ttl=127 time=0.534 ms
64 bytes from 10.0.4.15: icmp_seq=5 ttl=127 time=0.488 ms
64 bytes from 10.0.4.15: icmp_seq=6 ttl=127 time=0.555 ms
64 bytes from 10.0.4.15: icmp_seq=7 ttl=127 time=0.479 ms
64 bytes from 10.0.4.15: icmp_seq=8 ttl=127 time=0.502 ms
64 bytes from 10.0.4.15: icmp_seq=9 ttl=127 time=0.547 ms
64 bytes from 10.0.4.15: icmp_seq=10 ttl=127 time=0.466 ms
64 bytes from 10.0.4.15: icmp_seq=11 ttl=127 time=0.516 ms
64 bytes from 10.0.4.15: icmp_seq=12 ttl=127 time=0.591 ms
^C
--- 10.0.4.15 ping statistics ---
12 packets transmitted, 12 received, 0% packet loss, time 11264ms
rtt min/avg/max/mdev = 0.460/0.508/0.591/0.039 ms
[ec2-user@ip-10-0-2-163 ~]$
```

This is screenshot evidence of ICMP communication test from Web Server instance.

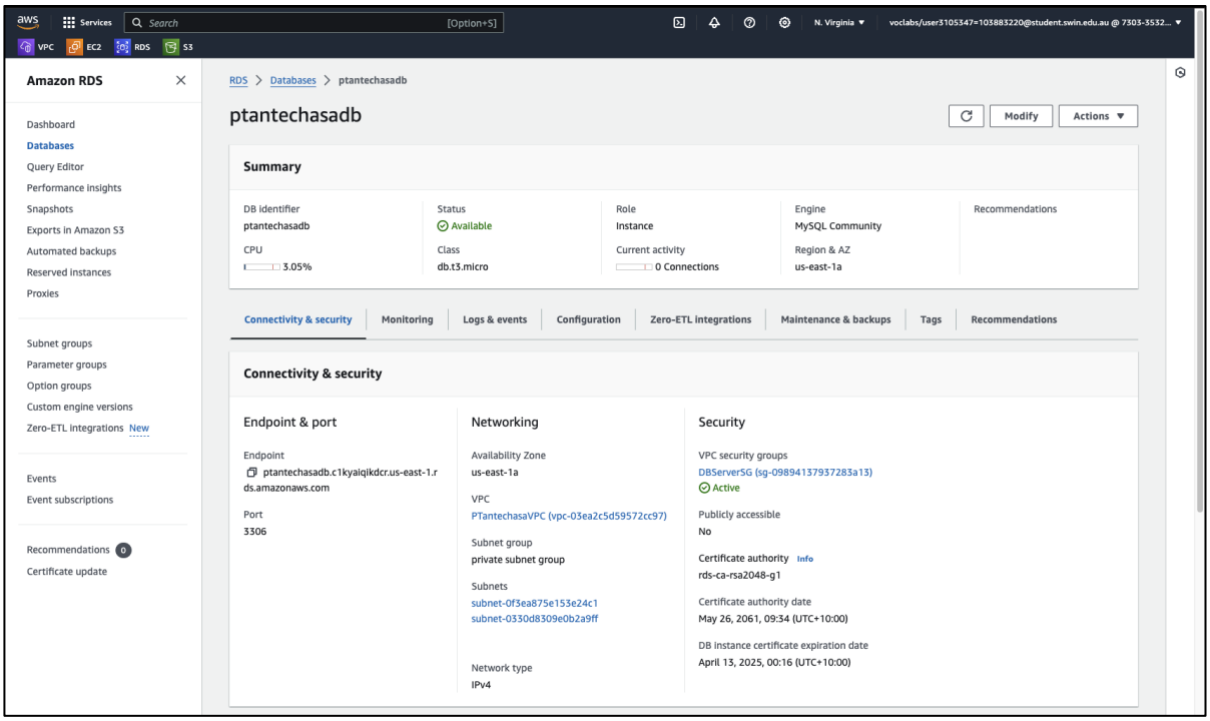
d. Test SSH Communication

```
ec2-user@ip-10-0-4-15:~
[ec2-user@ip-10-0-2-163 ~]$ chmod 400 "testkey.pem"
[ec2-user@ip-10-0-2-163 ~]$ ssh -i "testkey.pem" ec2-user@10.0.4.15
The authenticity of host '10.0.4.15 (10.0.4.15)' can't be established.
ECDSA key fingerprint is SHA256:WMe6clhB7E6djUjNtN/nn9YDRgZHR3mD5B3NJxS8nQo.
ECDSA key fingerprint is MD5:a6:b5:07:4a:70:cf:96:5e:a5:14:37:ae:9c:18:00:e8.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '10.0.4.15' (ECDSA) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-0-4-15 ~]$
```

This is screenshot evidence of SSH communication test from Web Server instance.

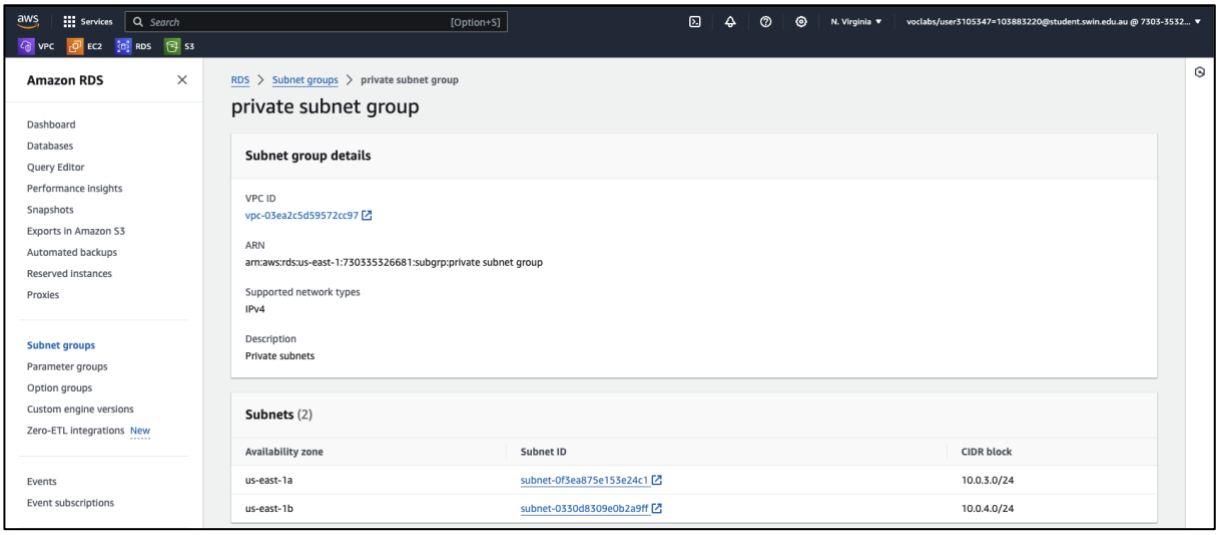
F. RDS

a. Database (Private Subnet 1)



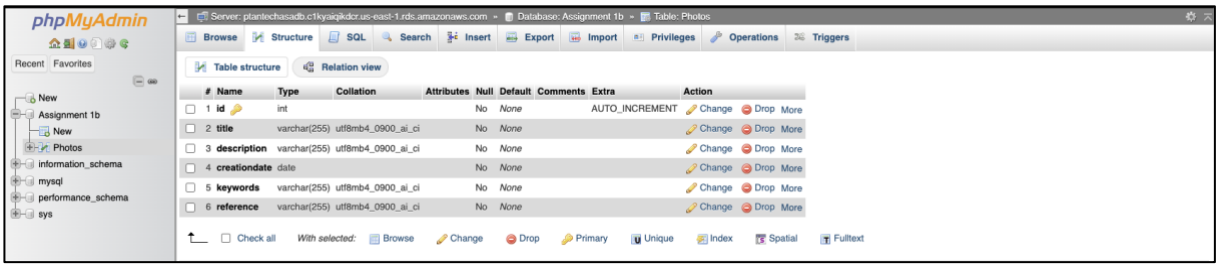
The establishment of this database involved the utilization of Amazon Web Services (AWS) Relational Database Service (RDS), which facilitated its creation and placement within Private Subnet 1, ensuring a secure and isolated network environment.

Private Subnet Group

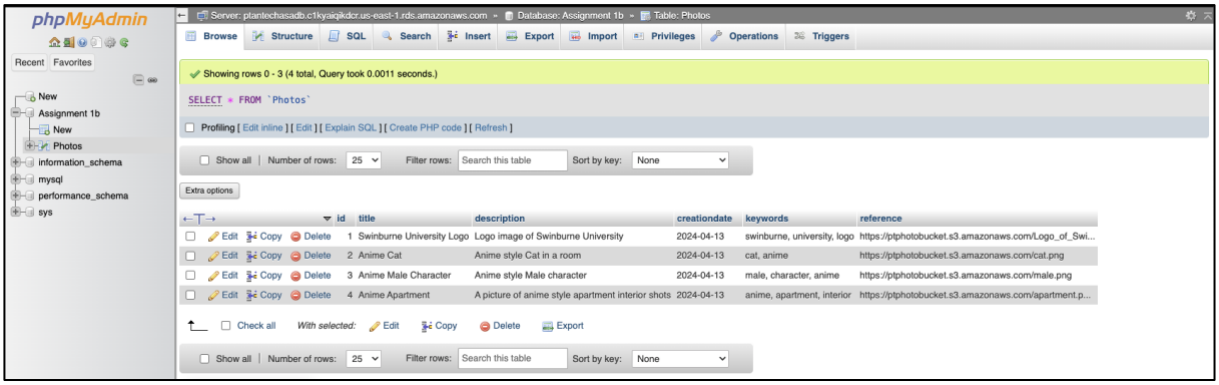


The database creation process involved the utilization of a private subnet group, ensuring that the database is situated within a network environment restricted from public access.

b. PhpMyAdmin

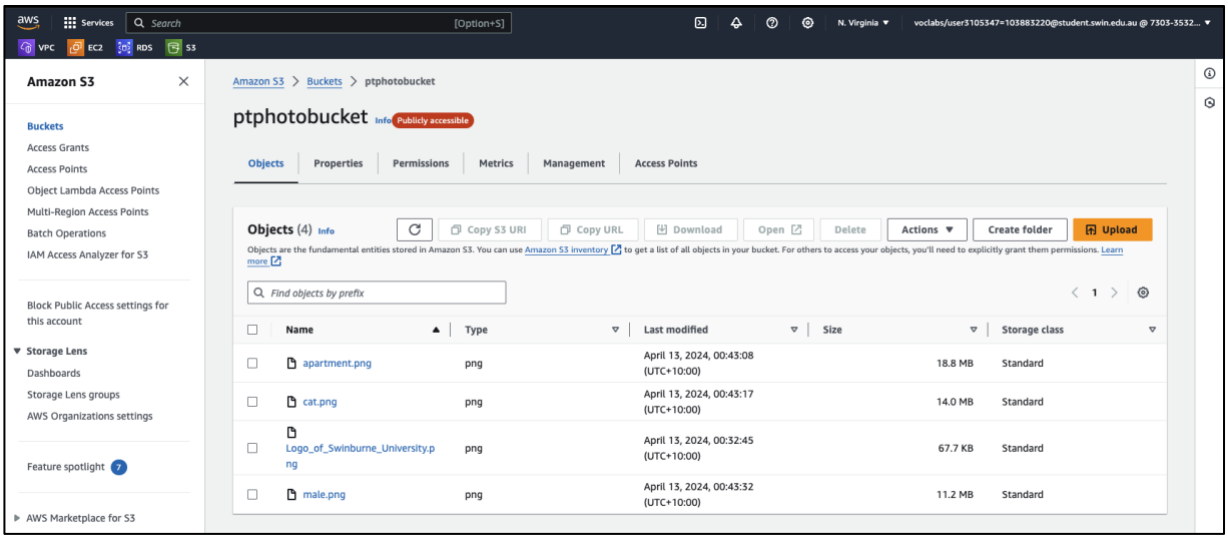


This is structure of "Photos" table within "Assignment 1b" database.

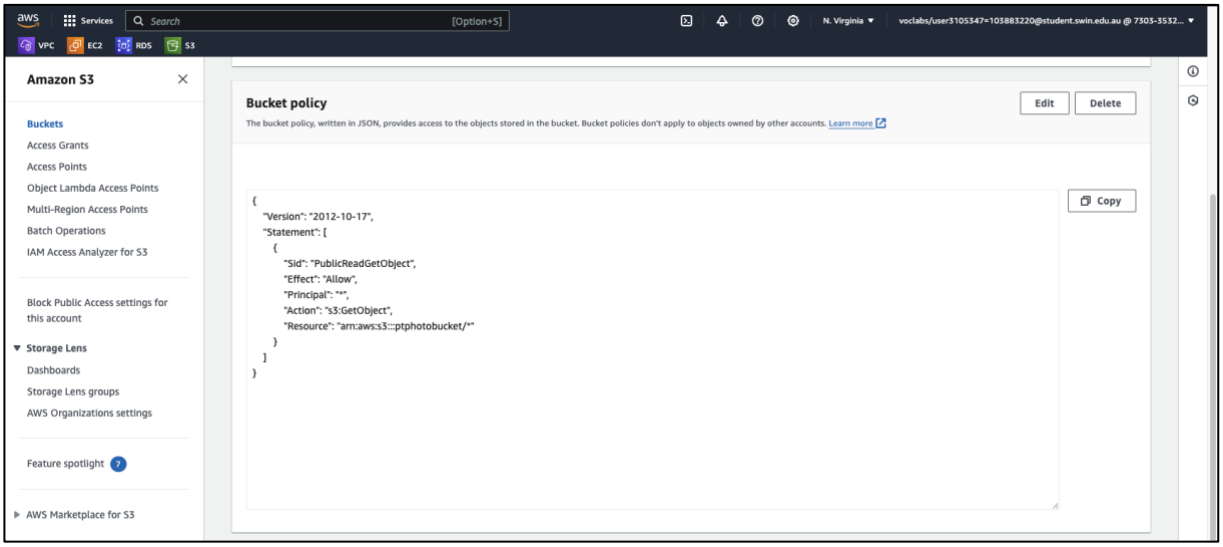


This is screenshot of meta-data that had been inserted to the database.

G. S3 Bucket



This is screenshot evidence of objects that's been uploaded to S3 bucket.



This is screenshot evidence of bucket policy which enable publicly access to the bucket.

III. FUNCTIONALITY

A. URL

http://54.83.150.188/cos20019/photoalbum/album.php

B. Web Page (album.php)

Student name: Pattarapol Tantechasa

Student ID: 103883220

Tutorial session: Tuesday 10:30AM

Uploaded photos:

Photo	Name	Description	Creation date	Keywords
	Swinburne University Logo	Logo image of Swinburne University	2024-04-13	swinburne, university, logo
	Anime Cat	Anime style Cat in a room	2024-04-13	cat, anime
	Anime Male Character	Anime style Male character	2024-04-13	male, character, anime
	Anime Apartment	A picture of anime style apartment interior shots	2024-04-13	anime, apartment, interior